

Light: Unit Review

Year 9 Science · end-of-year exam

Handout

The one idea

Light travels in **straight lines** from a source. When it meets something it is **reflected**, **absorbed**, **transmitted** or **refracted**. Every question in this unit asks two things: *what happens to the light*, and *where does it go next*? Draw the ray, then explain it.

1 How we see

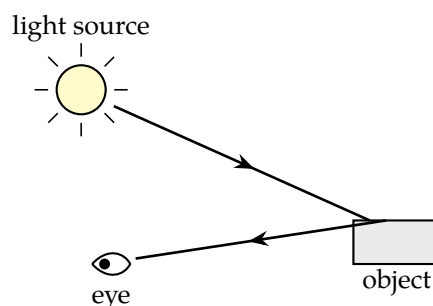
A **light source** gives out (emits) its own light: the Sun, a candle flame, a light bulb, a glow-worm, a phone screen. A **light reflector** only reflects light that falls on it: the Moon, a planet, a mirror, this page.

We see an object when **light from it enters our eyes**. For a non-luminous object the light comes from a source, is reflected by the object, and travels to the eye.

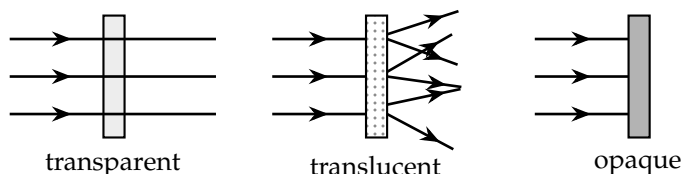
Write it in this order (*Why can we see the Moon?*)

1. The Sun **emits** light.
2. The light travels in straight lines to the Moon.
3. The Moon **reflects** some of it.
4. The reflected light travels to Earth and **enters our eyes**.

The Moon looks dimmer in daytime because the sky is already bright with scattered sunlight; it is **not** a light source.



2 Transparent, translucent, opaque



Word	What the light does	Examples
transparent	almost all light is transmitted straight through; objects behind are seen clearly	window glass, clear water, air
translucent	light passes through but is scattered ; objects behind look blurred	baking paper, frosted glass, milk bottle
opaque	no light passes through; it is absorbed or reflected	wood, wall, cardboard, jumper

3 Properties of light

1. Light travels in **straight lines**. Evidence: sharp shadows, a laser beam in smoke, a candle seen only through a *straight* tube.
2. Light travels **very fast**: 300 000 km/s in air or a vacuum.
3. Light can travel through a **vacuum** (empty space), e.g. from the Sun to Earth. Sound cannot: it needs particles.

Light slows down in materials (e.g. about 200 000 km/s in glass, 125 000 km/s in diamond) but is still very fast. This change of speed causes refraction (Section 7).

Worked example. The circumference of the Earth is 40 000 km. How many times could light go around the Earth in one second?

Step 1: in one second light travels 300 000 km.

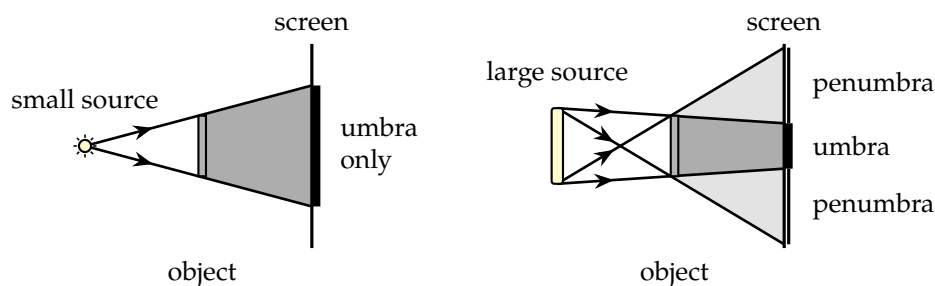
Step 2: number of times = $\frac{300\,000\text{ km}}{40\,000\text{ km}} = 7.5$ times.

Check: $7.5 \times 40\,000 = 300\,000$. ✓

Why we see lightning before we hear thunder: both happen together, but light (300 000 km/s) travels about a million times faster than sound (about 0.34 km/s), so the light reaches us first.

4 Shadows

A **shadow** forms where an opaque object **blocks** light. Because light travels in straight lines, the shadow has a similar **shape** to the object.



- **umbra:** the dark, sharp part where **all** the light is blocked.
- **penumbra:** the lighter, fuzzy edge where only **some** of the light is blocked. It forms only when the source is **large** compared with the object.

Situation	What you see	Why
small (point) source	sharp shadow, umbra only	every ray from the source is either blocked or not
large source	umbra with a penumbra edge	near the edge, part of the source is still visible
object moved closer to the source	shadow gets bigger	the edge rays spread out more before the screen

Drawing a shadow: use a ruler; draw straight rays from the *edges* of the source past the *edges* of the object to the screen; add arrows.

5 Colour

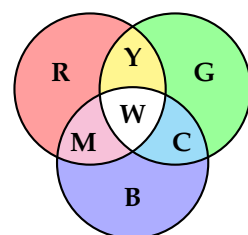
White light and the spectrum

White light is a **mixture** of the colours of the visible spectrum: **red, orange, yellow, green, blue, indigo, violet** (*Richard Of York Gave Battle In Vain*). A prism or a diffraction grating splits white light into these colours (Section 8).

Primary colours of light

The eye has three types of **cone** cell, sensitive to **red, green and blue** light. These are the **primary colours of light**; they cannot be made by mixing other colours. Mixing two primary colours gives a **secondary colour**; all three give **white**. Phone and TV screens use tiny red, green and blue **pixels**.

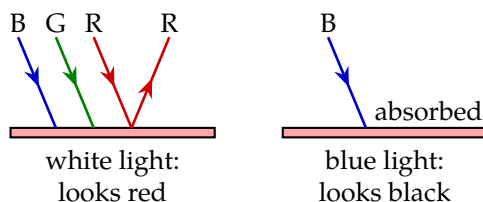
Careful: red, yellow, blue are primary colours of *paint* (pigment). Paint mixing is a different process.



R red, G green, B blue
 Y yellow = R + G
 M magenta = R + B
 C cyan = G + B
 W white = R + G + B

Coloured objects

An object **reflects** its own colour and **absorbs** all the others. A white object reflects all colours; a black object absorbs all colours. If the light shining on it contains **none** of the object's colour, nothing is reflected and the object looks **black**.

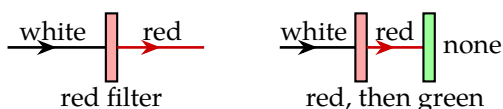


Write it in this order (*Why does a red apple look black in blue light?*)

1. The apple is red, so it **reflects only red** light and **absorbs** all other colours.
2. Blue light contains **no red** light.
3. The apple **absorbs** the blue light, so **no light is reflected** into our eyes.
4. With no light from it reaching our eyes, the apple looks **black**.

Filters

A filter **transmits** (lets through) only its own colour and **absorbs** the rest. A red filter in white light lets out red light only. A red filter followed by a green filter lets out **no light**.



Object colour	Light shone on it	Object appears
white	red	red (reflects the red it receives)
red	red	red
red	green or blue	black (nothing to reflect)
blue	white	blue
green	magenta (red + blue)	black
yellow	red	red (yellow objects reflect red and green)

6 Reflection

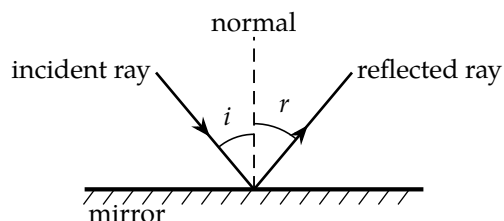
- **normal**: a line drawn at 90° to the mirror where the ray hits it (dashed).
- **angle of incidence** (i): between the incident ray and the **normal**.
- **angle of reflection** (r): between the reflected ray and the **normal**.

Law of reflection: the angle of incidence **equals** the angle of reflection: $i = r$.

Worked example. A ray hits a mirror at 25° to the mirror surface. Find i and r .

Step 1: angles are measured from the normal, which is at 90° to the mirror.

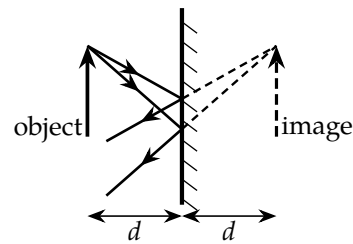
Step 2: $i = 90^\circ - 25^\circ = 65^\circ$, so $r = 65^\circ$.



The image in a plane mirror

The image in a plane (flat) mirror is:

1. the **same size** as the object;
2. the **same distance behind** the mirror as the object is in front;
3. **upright** (the right way up);
4. **laterally inverted** (left and right swapped);
5. **virtual**: the light only *appears* to come from it; it cannot be formed on a screen.



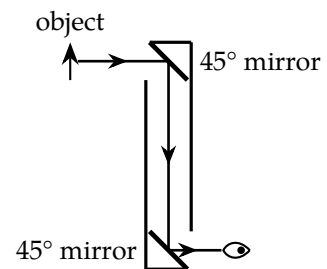
To locate an image: (1) measure the object's distance from the mirror and mark the image the same distance behind, on a line at 90° to the mirror; (2) draw two rays from the object to the mirror; (3) draw the reflected rays so that, extended backwards as **dotted lines**, they meet at the image.

	Real image	Virtual image
Meaning	light rays actually meet there	light rays only appear to come from there
On a screen?	yes	no
Examples	projector, camera, image on the retina	plane mirror, magnifying glass

Lateral inversion is why AMBULANCE is printed back-to-front on the front of the van: a driver ahead sees it the right way round in the rear-view mirror.

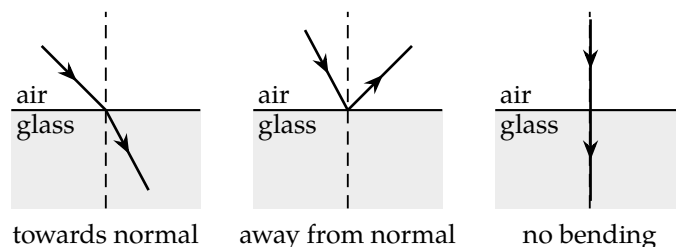


Uses of plane mirrors: a **periscope** has two parallel mirrors at 45° ; light turns through 90° at each mirror, so you can see over or around obstacles. Kaleidoscopes, dentist's mirrors and rear-view mirrors also use reflection.



7 Refraction

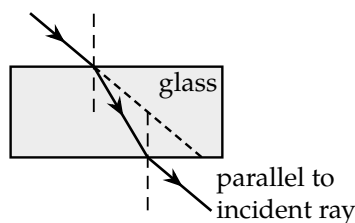
Refraction is the **bending** of light as it passes from one medium into another, because its **speed changes**. A medium in which light is slower is more **optically dense**.



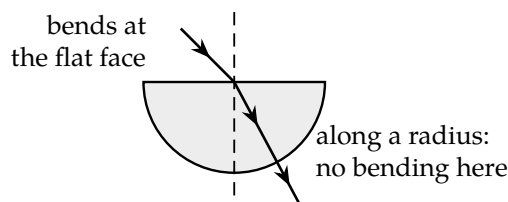
Into a slower medium (air $\rightarrow$ glass or water)	Into a faster medium (glass or water $\rightarrow$ air)	Along the normal
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slows down, bends **towards** the normal speeds up, bends **away from** the normal speed changes but **no bending**

Memory aid from the notes: *Strawberries Taste Nice*: **Slower** $\rightarrow$ **Towards** the Normal.



Rectangular block: the ray bends towards the normal going in and away coming out. It leaves **parallel** to the incident ray but shifted sideways.

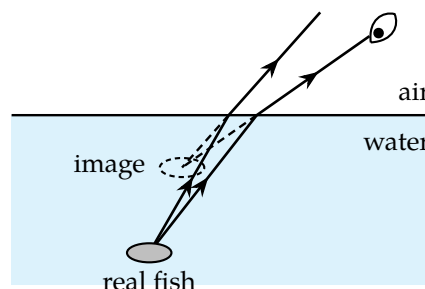


Semicircular block: a ray along a radius meets the curved face along the normal, so it does not bend there; it bends only at the flat face.

Refraction in everyday life: things under water look closer

Write it in this order (*Why does a fish look closer to the surface than it is?*)

1. Light from the fish travels from water into air.
2. It **speeds up**, so it bends **away from the normal** at the surface.
3. Our brain assumes light travels in **straight lines**, so it traces the rays straight back (dotted lines).
4. The rays appear to come from a point **higher** than the fish: we see a **virtual image** closer to the surface.

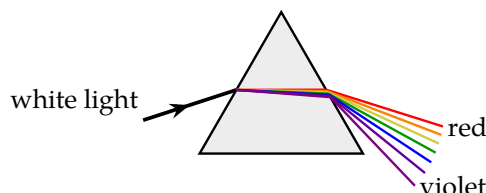


So a pool looks **shallower** than it is, a straw looks bent at the water surface, and a spear fisher must aim **below** where the fish appears.

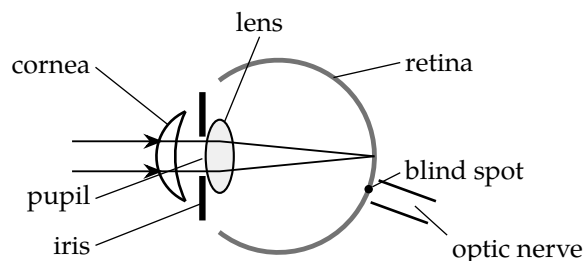
8 Dispersion

dispersion is the **splitting of white light** into its colours. In a prism, each colour travels at a **different speed**, so each is **refracted by a different amount**. **Violet bends the most; red bends the least**. A **rainbow** forms when sunlight is refracted and dispersed by raindrops.

The spread of colours in the diagram is exaggerated so it can be seen.



9 The eye



Part	Function
cornea	clear protective front; refracts (bends) the light most
iris	coloured muscle ; changes the size of the pupil
pupil	the hole that lets light in
lens	convex ; fine-focuses light on the retina; muscles make it fatter or thinner
retina	light-sensitive cells ; turns light into electrical signals
optic nerve	carries the signals to the brain
blind spot	where the optic nerve leaves: no light-sensitive cells

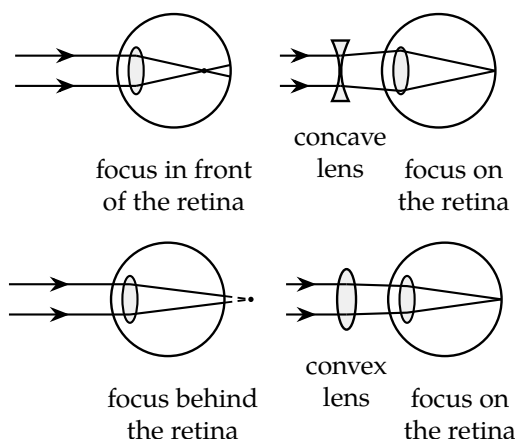
Pupil size: in **dim** light the iris makes the pupil **bigger** to let more light in; in **bright** light it makes the

pupil **smaller** to protect the retina.

Write it in this order (*How does the eye form an image?*)

1. Light from the object enters through the **cornea** and the **pupil**.
2. The cornea and the **convex lens** **refract** the light so the rays **converge**.
3. The rays meet on the **retina**, forming a **real**, upside-down image.
4. The retina's cells send electrical signals along the **optic nerve** to the **brain**, which turns the image the right way up.

Eye problems and how lenses correct them



	Sees clearly	The problem	Corrected with
short-sightedness (myopia)	near objects; far objects are blurred	rays converge too quickly : focus in front of the retina	concave (diverging) lens
long-sightedness (hypermetropia)	far objects; near objects are blurred	rays converge too slowly : focus behind the retina	convex (converging) lens

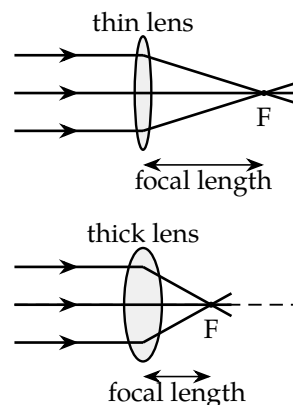
10 Lenses

A **convex** lens is fatter in the middle. It refracts parallel rays so they **converge** (come together) at the **focal point** F. The **focal length** is the distance from the centre of the lens to F. The ray through the centre of the lens does not bend.

A **thicker** (more curved) convex lens bends light **more**, so its focal point is **closer** to the lens (shorter focal length).

Magnifying glass in sunlight: rays from the Sun are parallel, so they converge at F; holding the paper at F concentrates the light energy on one small spot.

A **concave** lens is thinner in the middle. It makes parallel rays **diverge** (spread out).



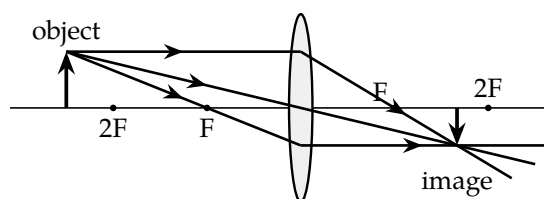
Beyond Year 9 · for interest only, not in the Year 9 exam

Refractive index. $n = \frac{\text{speed of light in a vacuum}}{\text{speed of light in the material}} = \frac{c}{v}$. Glass: $n = \frac{300\,000}{200\,000} = 1.5$; water 1.33; diamond 2.42.

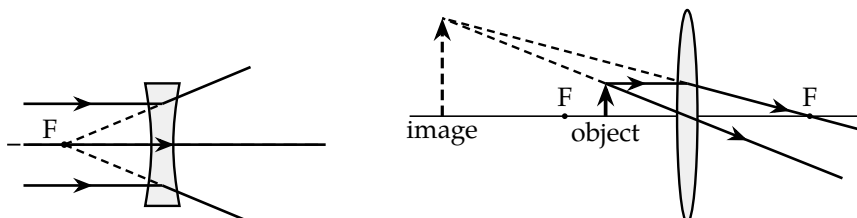
A larger n means slower light and more bending.

Snell's law. $n_1 \sin \theta_1 = n_2 \sin \theta_2$ (θ measured from the normal). Going into a material with a **larger** n , θ gets **smaller**: the ray bends towards the normal. This is the rule above, written as an equation.

Three special rays through a convex lens: (1) parallel to the axis $\rightarrow$ through F; (2) through the centre $\rightarrow$ straight on; (3) through F $\rightarrow$ parallel to the axis. Where they meet is the image. A concave lens spreads parallel rays out as if they came from F; a magnifying glass (object inside F) gives a virtual, upright, bigger image.



Object position (convex lens)	Image	Used in
beyond 2F	between F and 2F; real, inverted, smaller	eye, camera
at 2F	at 2F; real, inverted, same size	photocopier
between F and 2F	beyond 2F; real, inverted, bigger	projector
at F	no image (rays leave parallel)	spotlight
inside F	same side; virtual , upright, bigger	magnifying glass
any position (concave lens)	virtual, upright, smaller	glasses for myopia



11 Explain questions: Achieved, Merit, Excellence

Explain questions are marked on how far the answer goes. The same question, three levels:

Explain how a periscope lets you see over a wall.

- **Achieved** (states the idea): "It uses two mirrors to reflect the light."
- **Merit** (gives the reason): "Light from the object hits the top mirror and is reflected down the tube, then the bottom mirror reflects it into the eye. The mirrors are at 45° ."
- **Excellence** (links everything): "Light from the object above the wall hits the top mirror at an **angle of incidence of 45°** . By the **law of reflection** the angle of reflection is also 45° , so the light turns through 90° and travels down the tube. The bottom mirror is parallel, so the light turns through 90° again and travels horizontally **into the eye**. The eye receives light from an object it could not see directly." (A ray diagram with arrows and the normals supports it.)

The chain is always: **what the light does** → **why** (speed, absorbed, reflected, law) → **what we see** as a result.

12 Ways to lose marks

- L1 Angles measured from the mirror.** The angle of incidence and angle of reflection are measured from the **normal**, not from the mirror surface. A ray at 30° to the mirror has $i = 60^\circ$.
- L2 "The image is on the mirror".** A plane-mirror image is **behind** the mirror, the **same distance** behind as the object is in front. A person 2 m from a mirror is 4 m from their image.
- L3 Mixing light like paint.** Red + green *light* = **yellow**, not brown. Primary colours of light are **red, green, blue**; red, yellow, blue are paint (pigment) primaries.
- L4 "The filter adds colour".** A filter only **transmits** its own colour and **absorbs** the rest. It never turns white light red by "adding red".
- L5 Refraction direction reversed.** Into a slower (more optically dense) medium the ray bends **towards** the normal; out into a faster one it bends **away**. Along the normal: no bending.
- L6 "Light bends because it hits the glass".** Refraction happens because light **changes speed** at the boundary. An answer without "speed" does not reach Merit.
- L7 Missing arrows and dotted lines.** Rays need **arrows**; virtual rays (behind a mirror, back-extensions) are **dotted**. Drawn freehand or without a ruler loses the mark.
- L8 Pupil and iris swapped.** The **iris** (coloured muscle) changes the size of the **pupil** (the hole). "The pupil gets bigger by itself" is not enough.

These points come from the notes booklet and teaching observation, not from an examiner report.

13 Key words

Light source emits its own light.

Light reflector only reflects light.

Medium a substance light travels through.

Opaque / translucent / transparent blocks / scatters / transmits light.

Umbra full shadow; all light blocked.

Penumbra partial shadow; some light blocked.

Normal line at 90° to a surface.

Law of reflection $i = r$.

Plane mirror a flat mirror.

Real image can be formed on a screen.

Virtual image cannot be formed on a screen.

Laterally inverted left and right swapped.

Refraction bending of light caused by a change of speed between media.

Optically dense light travels more slowly in it.

Dispersion splitting white light into colours.

Spectrum red, orange, yellow, green, blue, indigo, violet.

Primary colours of light red, green, blue.

Secondary colours yellow, cyan, magenta.

Filter transmits its own colour, absorbs the rest.

Converge / diverge come together / spread out.

Focal point where parallel rays meet after a convex lens.

Focal length lens centre to focal point.

Myopia short-sighted; concave lens.

Hypermetropia long-sighted; convex lens.